

King County Real Estate Valuation & Pricing Analytics

Statistical property cleaning, luxury segment filtering, bedroom density analysis, and skewness modeling.

Dataset: kc_house_data.csv (21,613 Records)

Metrics: Price, Bedrooms, Bathrooms, Sqft

Tech: Pandas, NumPy, Matplotlib

1. Scenario Breakdown & Real Estate Metrics

Scenario 1: Data Loading & Basic Cleaning

Loaded 21,613 home sales. Imputed missing values: bedrooms with mode (3), bathrooms with mean (2.11), sqft_living with mean (2,079.9 sqft), and price with mean (\$540,088).

Scenario 2: Price Line Graph (First 10 Rows)

Subsampled the first 10 property transactions into NumPy arrays. Evaluated initial price dispersion from \$221,900 to \$687,500. Line plot: house_prices_line.png.

Scenario 3: Luxury Segment Filtering (Price > \$1,000,000)

Filtered homes priced > \$1,000,000 (1,465 luxury homes, 6.8%). The **4-bedroom (548)** and **3-bedroom (366)** formats represented the overwhelming majority of luxury inventory. Bar chart: expensive_houses_bar.png.

Scenario 4: Top 5 Bedroom Distribution Share

Quantified top bedroom configurations: 3 Bedrooms (45.5%), 4 Bedrooms (31.6%), 2 Bedrooms (12.8%), 5 Bedrooms (7.5%), 1 Bedroom (0.9%). Pie chart: bedroom_distribution.png.

Scenario 5: Advanced Tiering, NumPy Diff & Distribution Skewness

Created Price Category: *Affordable* (<\$500k: 12,558 homes, 58.1%), *Mid Range* (\$500k–\$1M: 7,590 homes, 35.1%), and *Luxury* (≥\$1M: 1,465 homes, 6.8%). Price distribution is heavily **right-skewed** with concentration in the affordable tier. Visuals: price_trend.png, price_category_stacked.png, price_histogram.png.

2. Visual Representation

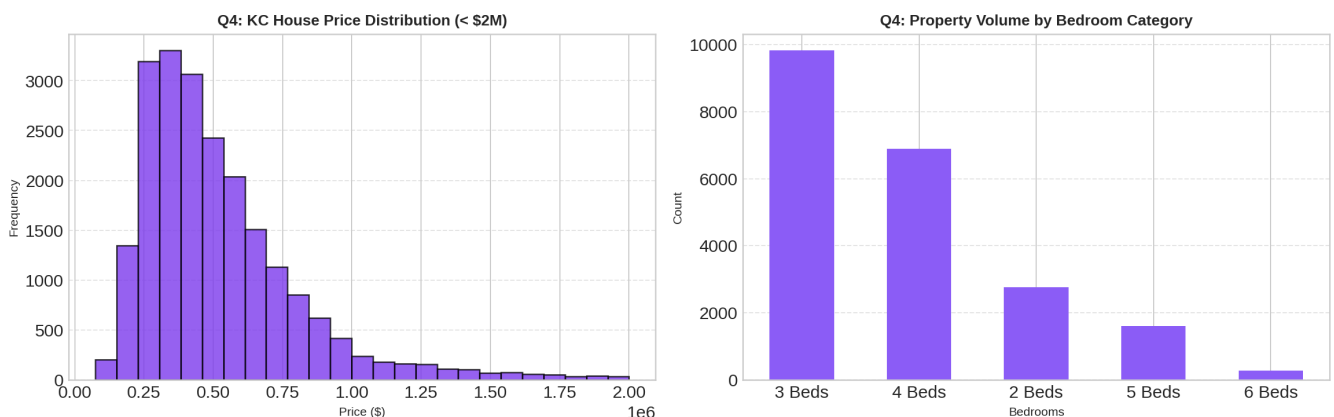


Figure 4.1: King County House Price Distribution (< \$2M) (Left) & Property Volume by Bedroom Tier (Right).

3. Python Code Snippet

```
import os, pandas as pd, numpy as np, matplotlib.pyplot as plt
df = pd.read_csv("kc_house_data.csv")
for c in ['bedrooms', 'bathrooms', 'sqft_living', 'price']:
    df[c] = pd.to_numeric(df[c], errors='coerce')
df['bedrooms'] = df['bedrooms'].fillna(df['bedrooms'].mode()[0])
df['price'] = df['price'].fillna(df['price'].mean())
diffs = np.diff(df['price'].to_numpy())
print(f"Luxury Homes Count (> $1M): {len(df[df['price']>1000000])}")
```